Maclaurin Series

Questions 1–42

Question 1. For each of the following, find the first 4 terms of the Maclaurin series and state the range of values for which the expansion is valid:

a) e^{3x}
b) e^{-2x}
c) $\ln(1+2x)$
d) $\ln(1-x)$
e) $\sin(2x)$
f) $\sin(-4x)$
g) $\cos\left(\frac{1}{2}x\right)$
h) $\cos(-x)$

Question 2. Write down the coefficient of the second term of the Maclaurin series expansion for each of the following:

a) $\cos\left(\frac{5}{2}x\right)$
b) e^{-4x}
c) $\ln\left(1+\frac{1}{2}x\right)$

Question 3.

a) Find the first three non-zero terms in the Maclaurin expansion of

$$\ln(3+x)$$

b) State the values for which the expansion is valid.

Question 4.

a) Find the series expansion in ascending powers of x as far as the term in x^3 for

$$\ln\left(1+\frac{1}{4}x\right)$$

b) State the values of x for which the expansion is valid.

c) Hence find the series expansion up to and including the term in x^3 for

$$\ln\sqrt{1+\frac{1}{4}x}$$

Question 5. Given that the coefficient of the third term in the Maclaurin series expansion of

$$\sin(px)$$

is $\frac{4}{15}$, find the value of p

Question 6. Given that the coefficient of the third term in the Maclaurin series expansion of

$$e^{px}$$

is 72, find the possible values of p

Question 7. Expand each of the following up to and including the x^3 term:

a) $\ln(4+4x)$
b) $\frac{3e^x+1}{e^x}$
c) $\ln(e-2x)$
d) $\frac{1}{x}\ln(1+x^2)$
e) $\frac{1}{x}e^{x^2}$
f) $\frac{1}{x}\ln(2-2x)$

Question 8.

a) Find and simplify the first five non-zero terms of the Maclaurin Series for $\cos(2x)$
b) Hence write down the first five non-zero terms of the Maclaurin Series for $\cos(-2x)$

Question 9.

a) Find and simplify the first four non-zero terms of the Maclaurin Series for $\sin(2x)$

b) Hence write down the first four non-zero terms of the Maclaurin Series for $\sin(-2x)$

Question 10. Find the first three non-zero terms of the Maclaurin series for each of the following, in each case stating the values of x for which the expansion is valid:

a) $(4+4x)^{\frac{1}{2}}$

b) $(4+x)^{\frac{1}{2}}$

c) $\frac{1}{\sqrt{4+5x}}$

Question 11. Find the series expansions, up to and including the term in x^3 of each of the following, in each case giving the range of values of x for which the expansion valid.

a) $\ln(3x^2 - x + 1)$

b) $\ln(2x^2 + 3x + 1)$

Question 12.

a) Find the Maclaurin series expansion, up to and including the term in x^3, of

$$\ln\left(1 + \frac{1}{2}x^2 + 4x\right)$$

b) Hence find the Maclaurin series expansion, up to and including the term in x^3, of

$$\ln(x^2 + 8x + 2)$$

Sums of Maclaurin series

Question 13. Find the first three non-zero terms of the Maclaurin series for each of the following:

a) $e^x + e^{-x}$
b) $\sin x + \cos x$
c) $\cos x - \sin x$
d) $e^x - \sin x$
e) $e^x - \cos x$
f) $2\sin 2x + 2\cos 2x$
g) $e^{ax} + e^{-ax}$
where a is a positive constant.
h) $\sin ax + x\cos ax$
where a is a positive constant.

Question 14. A function is given by

$$\mathrm{f}(x) = e^{ax} + (1+bx)^{\frac{1}{2}}$$

where a and b are positive constants.

a) State the range of values for which the expansion of $\mathrm{f}(x)$ is valid in terms of b
b) Find the first four terms in the expansion of $\mathrm{f}(x)$ in ascending powers of x, giving your answer in terms of a and b

It is given that the coefficient of x is -1 and the coefficient of x^2 is $\frac{5}{2}$

c) Find the values of a and b
d) Hence find the coefficient of x^3

Prerequisite knowledge for Q15-Q16: Hyperbolic Functions

Question 15. By considering the first six non-zero terms of the Maclaurin series for e^{2x} find the first three terms of the Maclaurin series for $\sinh(2x)$

Question 16. By considering the first five non-zero terms of the Maclaurin series for e^{2x} find the first three terms of the Maclaurin series for $\cosh(2x)$

Prerequisite knowledge for Q17-Q18: Hyperbolic Functions and Complex Numbers

Question 17. By considering the first five non-zero terms of the Maclaurin series for e^{ix}

a) find the first three non-zero terms of the Maclaurin series for

$$\cosh(ix)$$

b) Hence, state a trigonometric function which has the same first three terms.

Question 18. By considering the first six non-zero terms of the Maclaurin series for e^{ix}

a) find the first three terms of the Maclaurin series for

$$\sinh(ix)$$

It is known that $\sinh(ix) = i\mathrm{f}(x)$

b) State the function $\mathrm{f}(x)$

Question 19. A function is given by

$$\mathrm{f}(x) = k(1+2x)^{\frac{1}{3}} - \ln(1+5x)$$

It is known that when $\mathrm{f}(x)$ is expanded as a series in ascending powers of x that the second term has coefficient -1. Find:

a) the value of k
b) the series expansion in ascending powers of x as far as the term in x^3
c) the values for which the expansion $\mathrm{f}(x)$ is valid.

Question 20.

a) Fully factorise $1 - 2x - 8x^2$
b) Hence, find the first three non-zero terms of the Maclaurin series for

$$\ln(1 - 2x - 8x^2)$$

c) Determine the validity of

$$\ln(1 - 2x - 8x^2)$$

Products of Maclaurin series

Question 21. Find the first three non-zero terms of the Maclaurin series for each of the following:

a) $e^x \sin x$
b) $e^{-x} \sin x$
c) $\sin x \ln(1+x)$
d) $e^x \cos x$
e) $e^x \cos 2x$

Question 22.

a) Find the first four terms in the expansion, in ascending powers of x of

$$(3+5x)^{-1}$$

b) State the values for which the expansion of $(3+5x)^{-1}$ is valid.
c) Hence, find the first four terms in the expansion, in ascending powers of x of

$$\mathrm{f}(x) = \frac{\sin 2x}{3+5x}$$

d) State the values for which the expansion of $\mathrm{f}(x)$ is valid.

Question 23.

a) Find the Maclaurin expansion up to and including the x^5 term for both

$$\cos\theta \sin\theta \text{ and } \sin q\theta$$

where q is a positive constant.

b) Given that

$$\cos\theta \sin\theta = p \sin q\theta$$

where p and q are positive constants, find the value of p and q

Question 24. Given that the coefficient of the third term in the Maclaurin series expansion of

$$e^{px} \sin x$$

is $\frac{13}{3}$

Find the possible values of p

Question 25. Given that the coefficient of the third term in the Maclaurin series expansion of

$$e^x \sin px$$

is $-\frac{1}{3}$
Find the possible values of p

Question 26. Given that the coefficient of the third term in the Maclaurin series expansion of

$$\frac{\cos px}{e^x}$$

is -12
Find the possible values of p

Question 27. Find the values of a, b and k given that

$$\frac{\sin ax}{be^x} = \frac{1}{2}x - kx^2 - \frac{1}{12}x^3 + \cdots$$

where a, b and k are positive constants.

Using Maclaurin series

Question 28. Use the appropriate Maclaurin series to find the answer correct to 4 decimal places, for each of the following:

a) $e^{0.1}$
b) $\ln(0.9)$
c) $\sin\left(\frac{\pi}{18}\right)$
d) $\cos(14^\circ)$

Question 29. Samara is using a Maclaurin series to find a numerical value for $\ln 2.7$
Her workings are below:

$$\begin{aligned}\ln(1+1.7) &\approx 1.7 - \frac{1.7^2}{2} + \frac{1.7^3}{3} - \frac{1.7^4}{4} + \cdots \\ &\approx 1.7 - 1.445 + 1.6376\ldots - 2.088025 + \cdots \\ &= -0.1953 \text{ to 4 decimal places.}\end{aligned}$$

a) Use your calculator to find $\ln 2.7$
b) Explain why Samara's method isn't correct.
c) Given $\ln 3 \approx 1.09861$, use a Maclaurin series to find $\ln 2.7$ to 4 decimal places.

Question 30.
a) Find the series expansion of $\frac{1}{\sqrt{e^x}}$ up to and including the term in x^5
b) Hence find the value of $\frac{1}{\sqrt{e}}$ to 4 decimals places.

Question 31. By using a Maclaurin series up to the third non-zero term, find each of the following to two decimal places:

a) $\int_0^{0.5} e^x \, dx$

b) $\int_0^{0.5} \ln(1+x) \, dx$

c) $\int_0^{1.5} \sin x \, dx$

d) $\int_0^{1.5} \cos x \, dx$

e) $\int_0^{0.5} (1-x)^{\frac{1}{2}} \, dx$

Question 32. By using a Maclaurin series up to the x^3 term, find an approximation to

$$\int \frac{1}{\sqrt{1-x}} \, dx$$

Question 33.

a) Find a cubic approximation for

$$\ln(1+2x)$$

b) Hence use your approximation to find a non-zero approximate solution to

$$\ln(1+2x) = 8x^2$$

Give your answer to 3 significant figures.

c) Suggest an improvement to your approximation in part b) and justify your answer.

Question 34.

a) Find a quartic approximation for $\cos\theta$

b) Hence use your approximation to find an approximate positive acute solution to

$$\cos\theta = -\frac{1}{648}\theta^4$$

Give your answer to 3 significant figures.

Question 35. It is known that

$$\mathrm{f}(x) = e^x - \cos x$$

has a root near $x = 0$

a) Find the Maclaurin series expansion of $\mathrm{f}(x)$ up to and including the term in x^3

b) Hence find an estimate for the non-zero root of $\mathrm{f}(x) = 0$, that is the closest to 0, giving your answer to 3 significant figures.

Question 36. Using $x = 0.2$ in the expansion of

$$\ln\left(\frac{1+x}{1-x}\right)$$

a) find $\ln 1.5$ correct to 3 decimals places.

b) Find the percentage error of your solution to part a) with $\ln 1.5$ on your calculator.

Question 37.

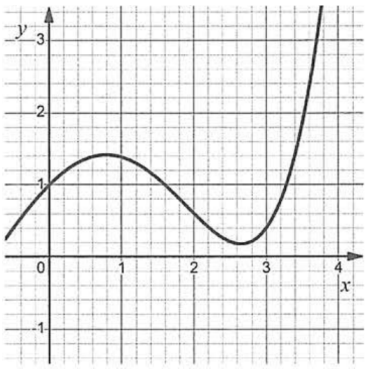
a) Find the first four terms in the expansion, in ascending powers of x of $e^x\sqrt{1+2x}$

b) Hence use $x = 0.5$ to find an approximation to $\sqrt{2}e^{\frac{1}{2}}$ giving your answer to 3 significant figures.

c) Find the percentage error between your answer in part b) and the exact value.

Question 38. By considering the Maclaurin expansion of both e^{-x} and $\cos(x)$ up to and including the x^2 term, sketch the graph of $y = e^{-x} - \cos x$ for $0 \leq x \leq 0.4$

Question 39. Two distinct functions f(x) and g(x) have been reduced to the first three terms of their Maclaurin series and f(x) + g(x) is shown in the graph below. The functions are two of e^x, $\cos x$, $\sin x$ and $\ln(1+x)$



Given that by including the fourth term of both the Maclaurin series for the curve is negative for values larger than $x \approx 2.3376$, identify the functions f(x) and g(x)

Question 40. A triangle ABC has sides of lengths $AB = 2$ and $AC = 3$, and the angle BAC is $2x$. Show that the area of ABC is approximately $px - qx^3$ where p and q are positive constants.
State the values of p and q

Question 41.

a) Find the series expansion in ascending powers of x as far as the term in x^3 for

$$\frac{\cos x}{\sqrt{1+x}}$$

b) Use your approximation in part a) to find an approximation of the equation of the tangent to the curve at $x = 0$ of $y = \frac{\cos x}{\sqrt{1+x}}$

Question 42.

a) Find the Maclaurin series of $\sin^2(2x)$ up to and including the term in x^4

b) Hence use your approximation to find a non-zero approximate solution to

$$\sin^2(2x) = 16x^3$$

Give your answer to 3 significant figures.